DevOps

Subject: Computer Science | NoteShare Library

DEVOPS

1. PRINCIPLES
Collaboration
Automation
Continuous improvement
Measure everything

2. CI/CD
Continuous Integration
Build, test on every commit
Continuous Delivery
Deploy to staging automatically
Continuous Deployment
Deploy to production automatically

Tools: Jenkins, GitLab CI, GitHub Actions, CircleCI

3. CONTAINERS
Docker:
Dockerfile
docker build, run, push
Docker Hub

Kubernetes:
Pods, Services, Deployments
ConfigMaps, Secrets
kubectl commands
Helm charts

4. INFRASTRUCTURE AS CODE
Terraform
Ansible
CloudFormation
Pulumi

5. MONITORING
Prometheus + Grafana
ELK Stack (Elasticsearch, Logstash, Kibana)
Datadog
New Relic

6. VERSION CONTROL
Git workflow
Feature branches
Code review
Semantic versioning

7. CONFIG MANAGEMENT
Ansible playbooks
Chef, Puppet
Environment variables